FFMI Medical Device Learning Templates

Learner-facing templates aligned to the integrated gaps and learning goals for FFMI Canvas medical device phases

How to use	Each template turns one or more identified course gaps into a concrete learner artifact. The left side of each template shows the instructional alignment; the right side is structured for learner completion.

Design basis. These templates were derived from the team's integrated gap analysis and learning goals, then organized as short templates, checklists, or decision aids that match the learner performances you identified.

Template overview

#	Template	Phase	Primary use
1	Device Concept Brief	P1	Translate need discovery into device-development terms.
2	Proof-of-Concept Plan	P1	Define the first feasibility signal and decision rule.
3	Benchtop Test Matrix	P2	Link requirements, tests, measures, and iteration decisions.
4	DHF-Style Traceability Log	P2	Track need → design decision → evidence across iterations.
5	Use-Related Risk Checklist	P2	Surface users, critical tasks, use errors, and harms.
6	Preclinical Evidence Rationale	P3	Justify what evidence is needed next and why.
7	Regulatory Pathway Decision Aid	P4	Compare a plausible 510(k), De Novo, or PMA path.
8	Adoption & Reimbursement Sketch	Cross	Clarify who pays, barriers, and needed value evidence.
9	Quality & Manufacturing Reality Check	Cross	Flag scale-up, QMS, design transfer, and manufacturability issues.

Template 1. Device Concept Brief

Aligned gap	Phase	Template type
Gap 1	Phase 1: Ideation / Discovery & Product Concept	Learner-facing tool
Learning-goal focus	**Design basis**	**Use**
• Foundational: Distinguish unmet need, device concept, intended use, indications for use, and context of use.	Adapted from the integrated Gap 1 learning goals and the team's commercialization timeline revision for Phase 1 (unmet need, device concept, intended use, context of use).	Complete during or after the related module as a short artifact.

Learner template

Field	Prompt / notes for completion
Unmet medical need	Who experiences the problem? What is happening now? Why do current solutions fall short?
Primary user(s) / stakeholder(s)	Who would use, influence, purchase, or be affected by the device?
Target population	Who is the intended patient or population? Include any obvious exclusions if known.
Use setting / context of use	Where would the device be used (clinic, hospital, home, lab, operating room, etc.)?
Device concept statement	In one or two sentences, what is the proposed device idea?
Intended use	What is the device intended to do?
Indications for use (preliminary)	Who is it for, or not for, and in what circumstances?
Current standard of care / alternatives	What is currently used to address the problem?
Preliminary value proposition	What value does the device add for the most important stakeholder?
Open questions for classification or pathway	What remains unclear that may affect risk level, evidence, or later regulatory planning?

Brief learner reflection (optional)

What remains most uncertain after completing this template?

Instructor note. This template is intentionally lightweight. It is meant to make the learner's reasoning visible rather than serve as a full regulatory, quality, or submission document.

Template 2. Proof-of-Concept (PoC) Plan

Aligned gap
Gap 2

Phase
Phase 1: Ideation / Discovery & Product Concept

Template type
Learner-facing tool

Learning-goal focus
• Foundational: Explain the difference between a proof of concept, an early prototype, and a final product.

Design basis
Adapted from the integrated Gap 2 learning goals and the commercialization timeline revision defining PoC as an early feasibility signal.

Use
Complete during or after the related module as a short artifact.

Learner template

Field	Prompt / notes for completion
Concept to be tested	What idea or mechanism is being explored?
Core feasibility assumption	What is the single most important assumption that must be tested first?
Why this assumption comes first	Why is this the highest-priority uncertainty at this stage?
Simple test or demonstration	What is the smallest practical test, simulation, or demonstration you will run?
Evidence to collect	What observations, measurements, or outputs will count as evidence?
Success criterion / decision rule	What result would suggest the concept shows enough promise to continue?
Failure interpretation	If the concept does not work, what would that mean?

Next step after the PoC	If promising, what is the next development action?

Brief learner reflection (optional)

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Template 3. Benchtop Test Matrix

Aligned gap
Gap 3

Phase
Phase 2: Design / Proof-of-Concept & Prototyping

Template type
Learner-facing tool

Learning-goal focus
• Applied: Identify one to three core performance requirements and translate them into simple, testable criteria.

Design basis
Adapted from the integrated Gap 3 learning goals and the commercialization timeline revision for early-stage benchtop testing.

Use
Complete during or after the related module as a short artifact.

Learner template

Requirement or function	Why it matters	Test method / setup	Measure or metric	Success criterion	Result / observation	Next design decision
[Enter item]						
[Enter item]						
[Enter item]						
[Enter item]						

Brief learner reflection (optional)

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Template 4. DHF-Style Traceability Log

Aligned gap
Gap 4

Phase
Phase 2: Design / Proof-of-Concept & Prototyping

Template type
Learner-facing tool

Learning-goal focus
• Foundational: Distinguish design input, design output, verification, validation, and design change.

Design basis
Adapted from the integrated Gap 4 learning goals and the commercialization timeline revision for DHF-style traceability.

Use
Complete during or after the related module as a short artifact.

Learner template

Need / requirement	Design input or criterion	Design output / decision	Verification or validation activity	Evidence or result	Design change / next action
[Enter item]					
[Enter item]					
[Enter item]					
[Enter item]					

Brief learner reflection (optional)

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Template 5. Use-Related Risk Checklist

Aligned gap
Gap 5

Phase
Phase 2: Design / Proof-of-Concept & Prototyping

Template type
Learner-facing tool

Learning-goal focus
• Applied: Identify intended users, use environments, and critical tasks; describe plausible use errors and harms.

Design basis
Adapted from the integrated Gap 5 learning goals and the commercialization timeline emphasis on user interaction and customer discovery during prototyping.

Use
Complete during or after the related module as a short artifact.

Learner template

Field	Prompt / notes for completion
Intended users	Who would directly use the device? Note any secondary users if relevant.
Use environments	Where will it be used? Consider constraints, distractions, or workflow conditions.
Critical tasks	What key user actions must be completed correctly for safe and effective use?
Plausible use error #1	What could go wrong in use?
Possible harm / consequence #1	What harm or downstream problem could follow?
Plausible use error #2	What else could go wrong in use?
Possible harm / consequence #2	What harm or downstream problem could follow?
Most important usability uncertainty to test next	What question should formative usability work answer?
How findings would inform design revision	What design feature, instruction, workflow element, or interface might change?

Brief learner reflection (optional)

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Template 6. Preclinical Evidence Rationale

Aligned gap
Gap 6

Phase
Phase 3: Pre-clinical Validation

Template type
Learner-facing tool

Learning-goal focus
• Foundational: Explain when biocompatibility, animal studies, risk management, and early endpoint planning may become necessary.

Design basis
Adapted from the integrated Gap 6 learning goals and the commercialization timeline revision for biocompatibility, animal rationale, residual risk, and early endpoints.

Use
Complete during or after the related module as a short artifact.

Learner template

Field	Prompt / notes for completion
Next safety or performance question	What must be answered next before further development?
Body contact / materials note	Does the device or its materials contact the body? If yes, how and for how long?
Biocompatibility implication	What does the body-contact profile suggest about needed evaluation or testing?
Need for animal model?	Is an animal study necessary, or can the question be addressed another way?
Reason an animal model would be justified	What question cannot be answered by bench or other evidence alone?
Main hazards and potential harms	What hazards are most relevant at this stage?
Risk controls already considered	What controls are planned or already in place?
Residual risk to document	What risk may remain after controls?
Preliminary endpoint(s)	What would be measured?
Measurement method	How would it be measured?
Why the endpoint matters later	How would this support later IDE or clinical planning?

Decision informed by this evidence	What go / revise / stop decision would the evidence support?

Brief learner reflection (optional)

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Template 7. Regulatory Pathway Decision Aid

Aligned gap
Gap 7

Phase
Phase 4: Clinical Studies & Regulatory Submission

Template type
Learner-facing tool

Learning-goal focus
• Foundational: Compare 510(k), De Novo, and PMA at a high level, including predicate logic and evidence expectations.

Design basis
Adapted from the integrated Gap 7 learning goals and the commercialization timeline revision for preliminary pathway selection.

Use
Complete during or after the related module as a short artifact.

Learner template

Decision factor	Current device idea	What this suggests about pathway
Risk level	[Describe preliminary risk level or class assumption]	
Intended use / indications	[What does the device do and for whom?]	
Potential predicate device	[Is there a similar legally marketed device?]	
Comparator logic	[What is similar, and what is meaningfully different?]	
Likely evidence burden	[Bench / preclinical / clinical expectations]	

Most plausible preliminary pathway	[510(k), De Novo, PMA, or unclear]	

Short written prompt

Short justification: Why is this pathway currently the most plausible, and what uncertainty might still change it?

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Template 8. Adoption & Reimbursement Sketch

Aligned gap
Gap 8

Phase
Cross-cutting: Market / Health System Adoption

Template type
Learner-facing tool

Learning-goal focus
• Applied: Identify the likely paying customer or procurement pathway and explain why that stakeholder influences commercialization success.

Design basis
Adapted from the integrated Gap 8 learning goals and the Canvas device roadmap language on paying customers, reimbursement, and procurement.

Use
Complete during or after the related module as a short artifact.

Learner template

Field	Prompt / notes for completion
Likely paying customer	Who is most likely to pay or approve purchase?
Procurement / payment pathway	Direct purchase, hospital procurement, reimbursement, payer pathway, or other?
Primary adoption stakeholder	Who inside the health system or payer environment most affects uptake?
Why this stakeholder matters more than the end user alone	What influence do they have on commercialization success?

Main adoption barrier	What barrier could slow adoption (workflow, cost, evidence, implementation, reimbursement uncertainty, etc.)?
Value claim that may reduce the barrier	What claim could make adoption more plausible?
Evidence needed to support that claim	What evidence would make the claim more credible?
Immediate next step	What should be investigated next?

Brief learner reflection (optional)

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Template 9. Quality & Manufacturing Reality Check

Aligned gap
Gap 9

Phase
Cross-cutting: Scale-up / Production Readiness

Template type
Learner-facing tool

Learning-goal focus
• Foundational: Explain what QMS expectations, design transfer, and manufacturing scale-up mean and why they influence planning before launch.

Design basis
Adapted from the integrated Gap 9 learning goals and the Canvas / timeline language on manufacturability, FDA-approved materials or processes, QMSR, and production realism.

Use
Complete during or after the related module as a short artifact.

Learner template

Assumption or issue	Why it matters	Possible effect on timeline	Possible effect on evidence / compliance	Possible commercialization risk / next action
[Example: material choice]				
[Example: manufacturing method]				

[Example: design transfer / documentation readiness]				

Short written prompt

Brief reflection: In simple terms, what QMS, design transfer, and scale-up concern feels most important for this concept right now?

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